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# Numerical and experimental investigations of the electrical and thermal performances of a novel PV thermal system

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ABSTRACT

Hybrid photovoltaic-thermal (PV-thermal) is a promising technology that could harness the maximum output from solar energy. In this work, a new thermal management system based on an innovative design of PV-thermal is designed and investigated under outdoor weather conditions in Sharjah, UAE. A numerical model is also developed to evaluate the thermal and electrical performances. Moreover, the energy and exergy evaluations are carried out to identify the performance enhancement of the PV system. The innovative absorber design adopted in this PV thermal system led to a considerable reduction of the module temperature. The results show that the novel cooling system enhanced both electrical and thermal performances. The maximum output power improved by an average of 15% with cooling, whereas the averaged module temperature decreased by almost 16 °C compared to the no-cooling case. The addition of front cooling to the PV-thermal system has also resulted in a further reduction of the module temperature by 26 °C and an increase of 17.6% in maximum output power with regard to the uncooled case. The exergy assessment showed that up to 61% increase in the overall exergy was achieved using the novel PV-thermal.

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## 1. Introduction

Solar technologies are gaining an increased attention to meet the energy requirements worldwide in terms of demand and environmental impacts. Both photovoltaic (PV) and solar thermal (ST) are the most mature solar technologies used to harness solar radiation. The solar PV systems are proliferating as they directly convert the sun rays into electricity using semiconductors at lower generation costs. Solar thermal systems are devices that harness the solar radiation and convert it to thermal energy, which can be directly useful in many purposes such as domestic hot water and space heating or indirectly used to generate electricity through a thermodynamic cycle.

Despite being one of the most competitive in the market with a lower cost of electricity, solar PV technologies suffer from low energy conversion efficiency, typically in the range of 4–26% [1]. One of the most important factors influencing the cell efficiency is the operating PV temperature which plays an important role in the energy conversion process. In fact, the operating temperature increase contributes to the decrease of the output voltage, fill factor, and power output of the PV cells [2]. Intending to overcome this challenge, it is necessary to cool down the PV cell temperature to improve its performance. Different methods were employed for thermal management of the PV modules and enhance their electrical efficiency [3–5]. One of the main approaches to decrease the working temperature of the PV cells and take full advantage of the collected solar radiation is to combine them within one solar collector in the so-called hybrid PV-thermal system. This concept is basically used to enhance the heat transfer between the PV module and the solar collector to maintain a lower PV temperature and increase the thermal yield of the solar collector.

The idea of the PV-thermal collector was initiated and documented as early as the mid-1970s [6]. The main origins of the PV-thermal are PV panels combined with thermal absorbers or heat exchangers [7]. In other words, the PV-thermal system combines the two main approaches by which solar irradiance can be utilized:

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solar electricity generation systems and solar thermal systems [8]. Both technologies manipulate the Sun's rays, in small-scale areas or broad solar farms. Hence, the rejected heat utilization and the improvement of the PV operational requirements boost up the overall conversion efficacy of the PV-thermal systems [9]. Martin Wolf [10] presented the pioneer study that introduced the flat-plate PV/T liquid-based system. From the 1980s, the PV-thermal was extensively studied, and much attention was paid to improving it and making it commercially viable [11]. Ramos et al. [12] stated that the PV-thermal collectors are a remarkably favourable technology, mainly when overall efficiency is of principal concern and/or the existing space is limited, such as in densely residential areas. Hence, the hybrid PV-thermal has the advantage of compact installation area compared to the two separated PV systems and flat thermal collectors [13]. Therefore, the PV-thermal system is considered one of the best candidates for a broad range of domestic and industrial purposes such as space heating and electricity production.

Moreover, the integration of the PV-thermal on building facades or rooftops gives these systems more credits and potentials. However, the PV-thermal technology still faces some challenges based on the available literature. One of the main challenges of PV-thermal is to propose a robust cooling system/thermal system to ensure the recommended operating conditions of the PV panels. The effective thermal system technique must expend a slight pumping requirement, with lower thermal resistance [14]. Furthermore, the PV-thermal system needs to reach a satisfactory uniform PV temperature in order to prevent current mismatching, hotspots and thermal stresses problems [15]. It is worth mentioning that the weight of the cooling device must be light, which is essential to reduce the tracking system requirements [16].

To achieve the desired enhancement in the PV-thermal system, numerous efforts were made and reported in the literature. Most of the published work considered modifying the PV-thermal module to improve the electrical, thermal, or both [17]. The methodology used to study this enhancement was based on experimental and numerical approaches. Sultan and Efzan [18] summarized the recent progress on the PV-thermal systems' advanced concepts and applications in a comprehensive review. The authors concluded that these systems have a remarkable potential to be a favourable technology although it requires a supplementary improvement in total efficiency and economic considerations. Kandeal et al. [19] categorized the modern cooling techniques according to the cooling method. The advances in PV-thermal systems' enhancement applying various cooling approaches was comprehensively reviewed and merged by Kandeal et al. Pang et al. [13] implemented a thorough assessment of PV-Thermal performance regarding the working fluid (WF), heatsink design, and terrestrial environments. The authors emphasized the influence of WF, feasible heatsink design, and the terrestrial condition. The authors also stated that cost consideration is of importance to more applicable and realistic applications.

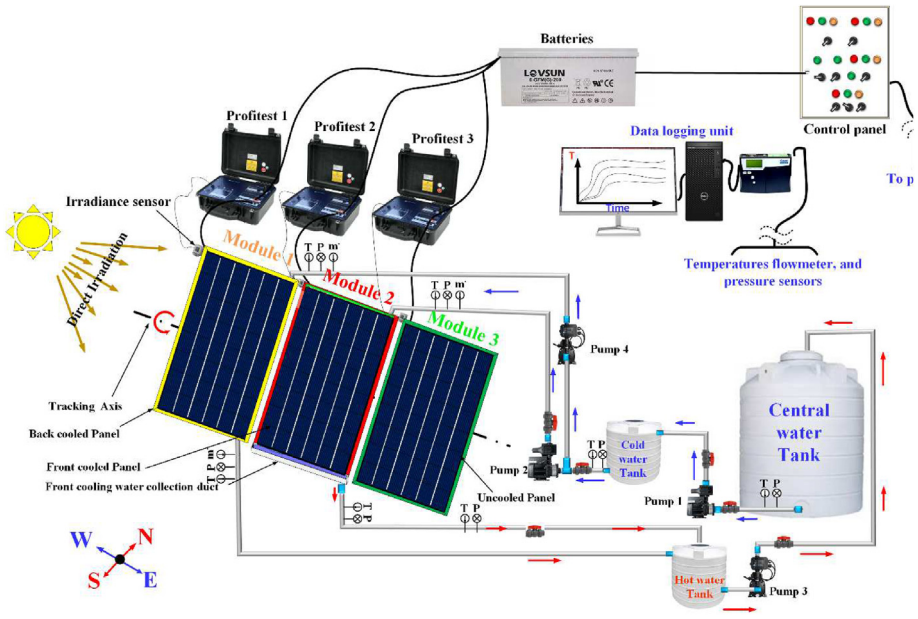
Various designs of PV-thermal collectors were proposed in the literature to seek a better electrical and thermal performance [20,21]. The conventional PV-thermal design is based on sheet and tube configuration, which integrates both technologies, i.e., solar PV and solar thermal, in their standard designs. However, this design presents a lower contact surface between the tubes and the absorber sheet as well as the high weight of the PV-T module due to the metal tube arrangements. Moreover, PV temperature uniformity is a leading factor in the performance of PV-thermal systems. Temperature non-uniformity ultimately influences PV lifetime, series resistance, and productivity.

Hence, offering a thermal system that can dissipate the excess heat from PV with a low-temperature gradient is an important challenge for the PV-thermal technology. Ahmed and Mohammed [22] experimentally used porous media on the PV-thermal performance. The thermal efficiency of up to 80.23% was achieved with the application of porous media and glazing. However, the presence of porous media was associated with an important increase in the pressure drop and pumping requirements. Yu et al. [23] proposed three different periodically grooved channels heat exchangers for the PV-thermal. The authors proposed these designs to prevent overheating, hotspots, and physical failure of the PV panels with low pressure drop. The CFD resources were used to assess the influence of groove designs on the PV-thermal performance. The results revealed that using expended grooves improved the flow distribution; consequently, the PV temperature gradient was decreased. However, the temperature contours reported by the authors show more than 20 °C differences within the PV panels, which could lead to thermal stress and current mismatching problems. The pressure drop also increased with the increase of groove distributions. Other designs were proposed, such as channel absorbers and roll-bond absorbers [24,25]. An innovative design of PV-thermal was manufactured by direct lamination of a special heat exchanger made of stainless steel [26,27]. The main idea of this design is to promote flow disturbance and heat transfer using built-in obstruction elements through the heat exchanger at the working fluid (WF) domains [28]. This PV-thermal showed interesting results for convective heat transfer coefficient between the heat exchanger and the absorber, and a low stagnation temperature which makes it more suitable for building integration applications. Only a few works were published using this innovative PV-thermal and simply tested under normal weather conditions. On the other hand, limited research work considered the performance of PV-thermal installed in very harsh weather conditions and high ambient temperatures, as the PV-thermal could be more degraded under high temperature and humidity levels [29].

The literature review indicates that only a few experimental works have been conducted to study new absorber designs under harsh weather conditions. In addition, none of the reported studies have evaluated the performance of the innovative PV-thermal collector using the special heat exchanger design under United Arab Emirates weather conditions from the point of view of first and second laws of thermodynamics. In this work, a new installation based on this innovative PV-thermal collector was designed and installed in Sharjah, United Arab Emirates (UAE). This installation is unique in situ installation that systematically includes various cooling strategies to compare these strategies under UAE weather conditions. In addition, a numerical model based on the energy balance of each domain was established to assess the thermal and electrical characteristics of the PV-thermal system. The experimental and numerical investigations were used to compare the novel PV-thermal system with a reference PV panel under the same weather conditions of Sharjah. Moreover, an exergy analysis is carried out to evaluate the exergy content of the innovative PV-thermal collector and compare it with the reference module. Based on the literature mentioned above and as far as the author's knowledge, no work has been reported on the proposed integrated system. Therefore, the authors sincerely believe that this work will be an essential inclusion in the literature and could help better understand the electrical, thermal, and thermodynamic characteristics of the innovative PV-thermal system.

## 2. Experimental setup

The experimental setup of the hybrid PV thermal collectors with various cooling strategies was designed and established on the rooftop of the University of Sharjah, UAE, as shown in Fig. 1 (a) and (b),. The installation consists of three identical PV modules (monocrystalline type and 280 Wp rated power). The hybrid PV




(a)



(b)

**Fig. 1.** (a) Schematic diagram of the thermal management system (b) different views of the actual experimental setup.

thermal configuration was adopted in 2 modules (module1 and 2), while one module (green dyed) was not equipped with active cooling for comparison purposes (module 3). A spraying system was attached to one hybrid PV thermal collector to test the front cooling technique where a water film is flowing on top of the PV module (red-dyed) to cool the front PV surface, as shown in Fig. 1 (a). The last module (yellow dyed) is cooled via the back side heat exchanger only. The electrical specifications of the used PV modules are summarized in Table 1. The thermal management system was designed in such a way that separates the cold fluid from the hot fluid as well as to maintain a low inlet temperature in a closed-loop operation. Therefore, a level control system was designed and installed on both tanks which were connected to a central tank to monitor the water level and temperature. The central tank was equipped with an external cooling system (using vapor compression cooling) to maintain the low temperature required in the cold tank. The details of the different layers within the hybrid PV thermal module are depicted in Fig. 2. The internal design of the heat exchanger promotes flow mixing and consequently improves the heat transfer. The proposed patterns also break up the cooling fluid into small passages which provides higher contact area and better temperature distribution. It is worth mentioning that the innovative PV thermal module is manufactured by direct lamination, where the flow field channel is sandwiched between two layers of stainless steel.

**Table 1**
Electrical specifications of the solar PV-thermal module.

| Parameter | Value |
|---|---|
| Type | Monocrystalline Solar Module |
| Pmax | 280 W |
| Output Tolerance | 0 ~ ± 3% |
| Vmpp | 31.95 V |
| Impp | 8.77 A |
| Voc | 38.88 V |
| Isc | 9.30 A |
| Test Conditions | 1000 W/m$^2$, AM 1.5, T = 25 °C |

A set of two batteries connected in series were used to store the electricity from the PV modules. A charge controller equipped with a maximum power point tracker (MPPT) was used to control and optimize the charge delivered from the PV module and dispatch it to the batteries. Water pumps were used in the installation to circulate the water in the loop. Different measurement devices were used to monitor the electrical and thermal characteristics of the designed setup. The description and precision of these devices are summarized in Table 2.

## 3. Uncertainty analysis

The assessment of measurement error is necessary to evaluate

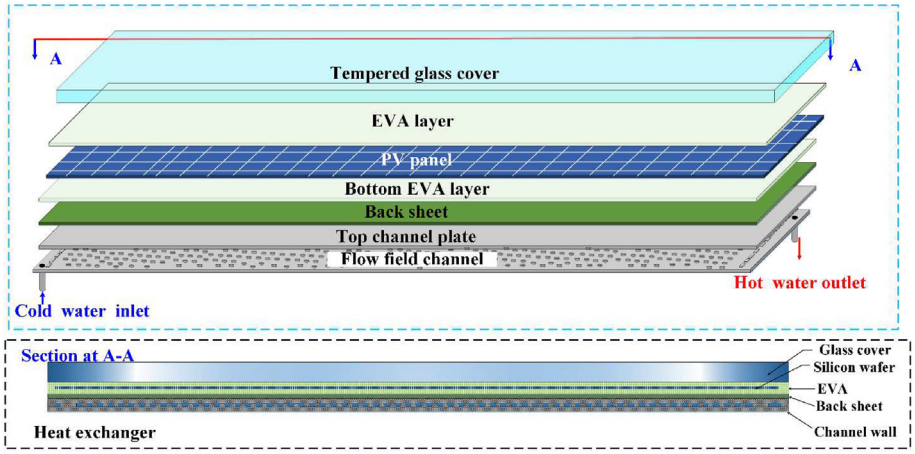



**Fig. 2.** Heat exchanger design and channel details.

**Table 2**
Specifications of measurement devices used in the present work.

| Instrument | Specifications |
|---|---|
| Pumps1 | Mini Pump XTL-3210-DC 12V<br>Max Pressure: ~1.0MPA<br>Max Flow rate: 6L/min |
| Power Supply | SDR-240-24 - DIN Rail Power Supply for pressure sensors and flowmeter |
| Flowmeter | TruFlo TK3 Series Paddle Wheel Flow Meters flow range: 0.3 m/s to 10 m/s<br>Accuracy: ± 1% of F.S |
| Pressure sensors | T2000 Pressure Transmitter<br>0–250 mbar (normal range)<br>Accuracy: ±0.5% F.S ±1.0% F.S |
| Thermocouples | PT100 Thermocouple Wire<br>Accuracy of ±0.35 °C |
| Portable Data Logger (Voltage, Temperature, pressure Recorder) | OMEGA: OM-SQ2040<br>Voltage accuracy: ±0.025% of reading + 0.005% of full scale<br>Temperature accuracy: ±0.05% of reading + 0.05% of full scale<br>Pressure accuracy: ±0.02% of reading + 0.015% of full scale |
| PROFITEST | Measurement of characteristic I-V curves at PV modules and strings up to 1000 V DC, 20 A DC, 20 kW |
| Batteries | Model: NRG-GFM-100 Ah<br>12 V, 200 AH Deep-cycle gel batteries |

the errors in experimental measurements. The total uncertainty ($\omega_x$) of a derivative measure of a quantity, $X$, due to various uncertainties ($\omega_1$,$\omega_2$, ...$\omega_n$) of the several determining quantities ($x_1$, $x_2$... $x_n$) were obtained using equation (1) [30]. The uncertainty analysis indicated that the maximum uncertainty values of measuring devices used in the current work are in acceptable ranges. As shown in Table 3 the maximum fractional uncertainty for all quantities is less than 5% in the current experimental work. Hence, the reliability of the measurement data in the current work is about 95% confidence.

$$\omega_X = \sqrt{\left(\frac{\partial X}{\partial x_1}\right)^2 \omega_{x_1}^2 + \left(\frac{\partial X}{\partial x_2}\right)^2 \omega_{x2}^2 + \cdots + \left(\frac{\partial X}{\partial x_n}\right)^2 \omega_{xn}^2} \quad (1)$$

**Table 3**
Maximum percentage uncertainty in the calculated and quantified quantities.

| Quaintly (unit) | Equation | Maximum uncertainty $\omega_x$/X (%) |
|---|---|---|
| $\eta_e$ (%) | Eq. (18) | ~0.04 |
| $P_{el}$ | Measured | ~0.50 |
| $E_{sol}$ | Eq. (25) | ~1.00 |
| $Ex_{el}$ | Measured | ~0.50 |
| $Ex_{Th}$ | Eq. (21) | ~4.84 |
| $Ex_{Total}$ | Eq. (22) | ~1.89 |

## 4. Mathematical model

A new numerical model was developed to evaluate the electrical and thermal characteristics of the hybrid PV-thermal system. The model is based on the first law of thermodynamics and follows the energy flow through the different parts of the system. Generally, solar radiation is transmitted to the solar cells of the PV-thermal system during sunny days. A portion of this solar radiation will be transformed to electrical power, and the other portion will be converted to heat that will be conducted through the back sheet and the heat exchanger, contributing to heat up the working fluid. The energy that reaches the PV-thermal system's backside will be lost to the environment through convective and radiative heat transfer. The thermal phenomenon of the system was modeled based on an energy balance model using the electrical/thermal analogy.

The following assumptions were applied in the current work:

- The heat capacity of the PV-thermal system was neglected while a quasi-steady state was considered.
- Heat was conducted in one dimension.
- A mean temperature was assumed across each layer of the PV-thermal system.
- The absorbed energy was transferred to the working fluid by forced convection heat transfer inside the heat exchanger.

- Transmissivity of the EVA is approximated to 100%.

The modeling approach used in the present work was based on the Hottel-Whiller-Bliss characteristic equation for flat plate collector (Eq (2)). This procedure could be extended to study other solar collector designs by deriving appropriate collector efficiency factor $F'$ and overall loss coefficient $U_L$ [31].

$$Q_u = F_R A_c [S - U_L(T_i - T_a)] \tag{2}$$

A schematic of the different layers and the equivalent thermal network for the PV-thermal system is shown in Fig. 3.

$U_t$, $U_1$, and $U_2$ represent the overall heat transfer coefficient from the PV cell to the surrounding through the glass cover, the overall heat transfer coefficient from the cell to the WF and the overall heat transfer coefficient from the WF to the ambient through the backside of the module, respectively. All these heat transfer coefficients include the different heat transfer modes (conduction, convection, and radiation) taking place between the different parts of the PV-thermal module, as shown in the following equations.

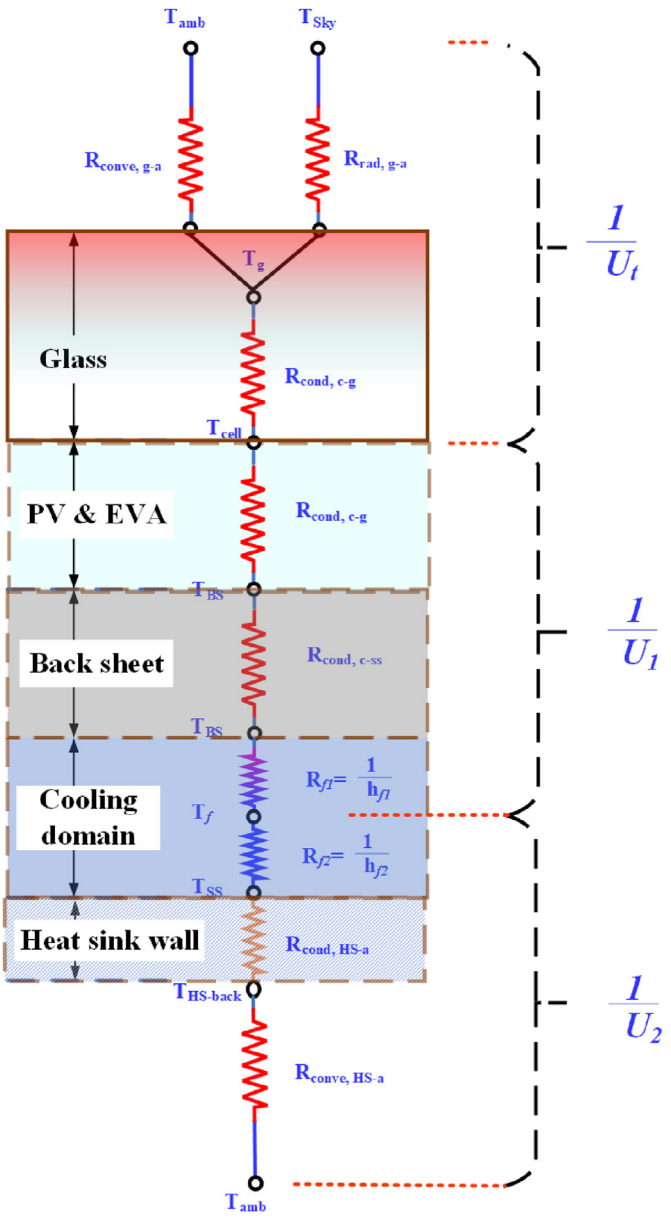



**Fig. 3.** Thermal resistance circuit diagram for the studied PV-thermal system.

$$U_t = \frac{1}{\left[h_{rad,g-s} + h_{conv,g-a}\right]^{-1}} + R_{cond,c-g} \tag{3}$$

where $h_{rad,g-s} h_{conv,g-a}$ $R_{cond,c-g}$ are the radiative heat transfer between the module and the sky, heat transfer by convection between the module and the ambient, and the conduction resistance between the cell and the glass cover, respectively.

$$U_1 = \left[R_{cond,c-g} + R_{cond,c-ss} + \frac{1}{h_{f1}}\right] \tag{4}$$

where $R_{cond,c-ss}$ and $h_{f1}$ are the conduction resistance between the cell to the stainless-steel sheet and the convective heat transfer coefficient between the WF and the upper stainless-steel sheet.

$$U_2 = \left[\frac{1}{U_b} + \frac{1}{h_{f2}}\right] \tag{5}$$

where $U_b$ and $h_{f2}$ are the overall back loss coefficient from the fluid to the ambient and the convection heat transfer coefficient between the fluid and the lower Stainless-Steel sheet. The heat transfer coefficients $h_{f1}$ and $h_{f2}$ between the WF and the stainless-steel sheets are evaluated based on the following correlations [32].

$$h_{f1} = 485.03 * \dot{V}^{0.1235} \tag{6}$$

$$h_{f2} = 465.73 * \dot{V}^{0.1394} \tag{7}$$

Energy balances on the solar cell and the WF provides the next equations:

$$S = U_t(T_c - T_c) + U_1\left(T_c - T_f\right) \tag{8}$$

$$q_u = U_t(T_c - T_a) + U_1\left(T_c - T_f\right) \tag{9}$$

Using Equation (3), the temperature of the solar cell could be defined as;

$$T_c = \frac{S + U_t T_a + U_1 T_f}{U_t + U_1} \tag{10}$$

After substituting the solar cell temperature, equation (3) could be rearranged as;

$$q_u = F'\left[S - U_L\left(T_f - T_a\right)\right] \tag{11}$$

where,

$$F' = \frac{U_1}{U_1 + U_t} \tag{12}$$

$$U_L = U_t + \frac{U_2}{U_1}(U_t + U_1) \tag{13}$$

$$S = (\tau\alpha)_{eff} G(1 - \eta_e) \tag{14}$$

The fluid temperature could be written in the direction of the flow

$$T_f(x) = \frac{S}{U_L} + T_a + \left(T_{fi} - T_a - \frac{S}{U_L}\right)\exp\left(-\frac{U_L F' x}{\dot{m}}\right) \quad (15)$$

The latter equation could be integrated over the width of the collector to obtain the mean fluid temperature.

$$T_{fm} = \frac{1}{W}\int_0^W T_f(x)dx \quad (16)$$

The design parameters of the hybrid PV-thermal system are summarized in Table 4.

The thermal efficiency of the hybrid PV-thermal collector is well-defined as the proportion of the effective thermal energy to the captured solar radiation.

$$\eta_{th} = \frac{Q_u}{GA} \quad (17)$$

The electrical efficiency of the PV was calculated using the temperature-dependent equation

$$\eta_e = \eta_r[1 - \beta(T_c - T_r)] \quad (18)$$

where, $\beta = 0.004K^{-1}$ is the temperature coefficient, $\eta_r$ is the PV module's reference efficiency, and $T_r$ is the reference temperature. These reference values were obtained from the technical specifications of the PV module under the Standard Test Conditions (Table 1).

## 5. Exergy analysis

Exergy analysis was performed to estimate the exergy losses and show the obtainable energy toward more efficient systems [33]. As the generated electrical energy ($E_{el}$) is practical, it can be considered pure exergy; consequently, electrical exergy ($Ex_{el}$) is equivalent to $E_{el}$ [34]. On the other hand, the thermal exergy ($Ex_{th}$) is not the same as thermal power. Two main approaches are commonly used in the literature to evaluate the thermal exergy in case of PV-thermal collector. The first approach is based on the thermal power extracted by the fluid at the outlet temperature. The thermal exergy $Ex_{th}$ is evaluated using a Carnot factor considering the fluid outlet temperature as the heat source [1,2]:

$$Ex_{th} = \dot{m}_f C_p \underbrace{\left(T_{f,out} - T_{f,in}\right)}_{Thermal\,power}\underbrace{\left(1 - \frac{T_0}{T_{f,out}}\right)}_{Carnot\,factor} \quad (19)$$

where, $T_{f,in}$ $T_{f,out}$, and $T_0$ refer to the WF inlet temperature, WF outlet temperature and reference temperature, respectively.

However, the second approach considers the exergy change due to the fluid temperature change based on the Gouy-Stodola Theorem [35]. Following Evola and Marletta [36], the Carnot factor term is evaluated using this approach as a function of the WF mean temperature ($T_m$), which is defined as:

$$T_m = \frac{h_{out} - h_{in}}{S_{out} - S_{in}} = \frac{\left(T_{out} - T_{f,in}\right)}{\ln\left[\frac{T_{f,out}}{T_{f,in}}\right]} \quad (20)$$

where $(h_{out} - h_{in})$ and $(S_{out} - S_{in})$ describe the enthalpy and entropy differences over the WF from, respectively. Hence, the thermal exergy $Ex_{th}$ using the second approach is calculated as the following:

$$Ex_{th} = \dot{m}_f C_p \underbrace{\left(T_{f,out} - T_{f,in}\right)}_{Thermal\,power}\underbrace{\left(1 - \frac{T_0}{T_m}\right)}_{Carnot\,factor} \quad (21)$$

This approach is more appropriate for determining the exergy content of liquid flow encountering in the heating process from inlet to outlet temperature. In this work, the second approach is adopted, and Equation (21) is used to calculate the thermal exergy.

The total exergy is equal to the sum of $E_{el}$ and $E_{th}$ as follow:

$$E_{tot} = Ex_{th} + Ex_{el} \quad (22)$$

The exergy losses of PV-thermal can be determined based on the subsequent equation:

$$E_{loss} = E_{sol} - Ex_{th} - Ex_{el} \quad (23)$$

The exergy is a measure of the maximum possible work in a given environment. In order to evaluate the solar radiation exergy, different methods have been proposed for solar applications. Among these methods, the equations proposed by Petela [37] and Jeter [38] (Equations (24) and (25)) are the most commonly used in the literature [39]:

$$E_{sol} = \left[1 - \frac{4}{3}\left(\frac{T_0}{T_{Sun}}\right) + \frac{1}{3}\left(\frac{T_0}{T_{Sun}}\right)^4\right] \times G \quad (24)$$

$$E_{sol} = \left(1 - \frac{T_0}{T_{Sun}}\right) \times G \quad (25)$$

where $T_{sun}$ Sun surface temperature which is about 5800 K [39,40].

These methods accurately predict the solar exergy when the dead state (system reaches thermodynamic equilibrium with it surrounding) is associated with a blackbody converter for fully concentrated radiation [41]. However, in case of PV-thermal system the solar radiation is not concentrated, and the converter is not a blackbody but a photovoltaic converter (selective absorber). In such situation, Badescu [42] proposed a more rigorous and complicated relationship to determine the solar exergy for unconcentrated solar radiation associated with selective converters (such as photovoltaic

**Table 4**
Dimensions and parameters of the PV-thermal collector.

| Dimensions | Value | Parameters | Value |
|---|---|---|---|
| Length ($L$) | 1.67 m | Transmissivity of the glass | 0.99 |
| Width ($W$) | 0.99 m | Absorptivity of the solar cell | 0.8 |
| Thickness of the glass cover | 2 mm | Emissivity of the PV-T | 0.9 |
| Thickness of the solar cell | 0.6 mm | Conductivity of the glass cover | 1 W/mK |
| Thickness of the EVA | 0.6 mm | Conductivity of the solar cell | 130 W/mK |
| Thickness of the backsheet | 0.14 mm | Conductivity of the EVA | 0.23 W/mK |
| Thickness of the SS sheet | 0.6 mm | Conductivity of the backsheet | 0.2 W/mK |
| Thickness of the water layer | 2 mm | Conductivity of the SS sheet | 025 W/mK |

cell). This relation considers the effect of the band gap energy and voltage across the solar cell on the exergetic factor. However, for low values of the cell to Sun temperatures (such for PV-thermal system), Petela and Jeter relations could be a good approximation [42,43]. A comparison between the three methods under the studied conditions (monocrystalline solar cell, same environment temperature and typical cell temperature) is conducted at different radiation temperatures as shown in Fig. 4. The results show a small difference between the three methods with an error of less than 3% (at Sun radiant temperature). For the sake of simplicity, the equation of Jeter (Equation (25)) is adopted in the exergy analysis.

## 6. Results and discussion

The PV-thermal system was tested on 18/04/2020 at the University of Sharjah, Sharjah UAE. The panel was tilted by 15° to optimize the solar irradiance received on the solar modules. The hourly weather conditions are summarized in Table 5. A maximum irradiance of 924 W/m$^2$ was recorded at 11 a.m. whereas the highest ambient temperature was found at 13h with a value of 33.9 °C.

The I-V curves are obtained using characteristic curve tracer Profitest, which include an irradiance sensor and temperature sensors. The accuracy of the instrument used in the experiment is given in Table 2.

The numerical and experimental results of both PV reference and PV-thermal module with back cooling are compared under the same climatic conditions. In order to evaluate the deviation between the measured values and numerical findings, the coefficient of correlation ($r$) and root mean square percent deviation ($e$) are calculated using the following expressions

$$e = \sqrt{\frac{\sum \left[100 \times (X_{sim,i} - X_{\exp,i})/X_{sim,i}\right]^2}{n}} \tag{30}$$

$$r = \frac{n\left(\sum X_{\exp} \times X_{sim}\right) - \left(\sum X_{\exp}\right) \times \left(\sum X_{sim}\right)}{\sqrt{n\left(\sum X^{\exp} 2\right) - \left(\sum X_{\exp}\right)^2} \times \sqrt{n\left(\sum X_{sim}^2\right) - \left(\sum X_{sim}\right)^2}} \tag{31}$$

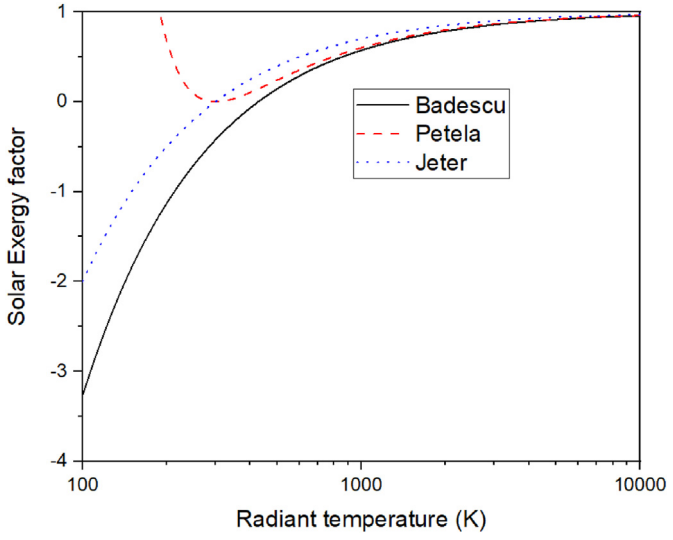



**Fig. 4.** Solar exergy factor versus different radiation temperatures using three methods: Badescu, Petela, and Jeter.

**Table 5**
Hourly weather conditions of 18th of April.

| Time (h) | Irradiance (W/m$^2$) | Ambient Temperature (°C) | Wind speed (m/s) |
|---|---|---|---|
| 9 | 808 | 29.9 | 0 |
| 10 | 875 | 31.6 | 0.1 |
| 11 | 924 | 33.8 | 0.2 |
| 12 | 903 | 33.5 | 0.5 |
| 13 | 815 | 33.9 | 0.7 |
| 14 | 625 | 32.9 | 1.0 |
| 15 | 479 | 33.2 | 0.8 |

Fig. 5 (a) compares the module temperature and maximum power between the reference PV module and the numerical model without cooling. A good matching between mathematical and experimental findings was obtained with correlation relation $r = 0.96$, and root mean square deviation $e = 2.79\%$ for module temperature. Moreover, electrical performance was also compared where the maximum measured power matches the numerical results with $r = 0.99$, and $e = 2.26\%$.

The impact of active cooling is also studied by comparing the thermal and electrical performances of the experimental and numerical results of the PV-thermal system, as shown in Fig. 5 (b). As expected, both electrical and thermal performances have improved with cooling. It is then observed that the daily maximum output power increased by an average of 15% with backside cooling, whereas the averaged module temperature decreased by almost 16 °C compared to the no-cooling case. When comparing the numerical and experimental results, a reasonable deviation was also obtained with correlation relation $r = 0.94$, and root mean square deviation $e = 8.14\%$. Despite this agreement, the numerical results underestimated the measured module temperature, probably due to the effect of the metallic structure that will contribute to heating up the module. This effect was not considered in the numerical model and could have an influence on the module temperature when cooling is activated. The maximum power comparison shows a good agreement between the numerical and experimental results with $r = 0.99$ and $e = 1.05\%$.

The working fluid used for numerical and experimental investigations is water at a temperature of 25 °C and a flow rate of 3 L/min. The results of the outlet fluid temperature are shown in Fig. 6, where a good agreement is observed with correlation relation $r = 0.95$ and root mean square deviation $e = 2.3\%$. These errors are within the accuracy of the primary and derived quantities of experimental value in the measurements as discussed in detail in section 3. The WF temperature had the same tendency as the module temperature but with lower magnitude.

Furthermore, the PV-thermal installation includes a PV-thermal module equipped with a front cooling method where WF was pumped at a rate of 1.2 L/min through the spraying system. This module was tested using both cooling methods, i.e., front, and back, at the same time. The surface temperature and output power results for the three solar modules are depicted in Fig. 7.

As expected, the module with double cooling showed the highest output power and lower module temperature compared to the other modules. The temperature of the PV module with double cooling has decreased by an average of 26.8 °C, whereas the output power increased by an average of 17.6% with respect to the reference module. The effective cooling using the innovative heat exchanger on the back of the PV module has significantly reduced cell temperature. This reduction improved the electrical efficiency of the PV module by an average of 14.3% with the back cooling, as shown in Fig. 8. The addition of the front cooling also contributed to cool down the module temperature further. As a result, the electrical efficiency was increased by an average of 18.3% with respect to the reference module. It is worth noting that the back cooling for

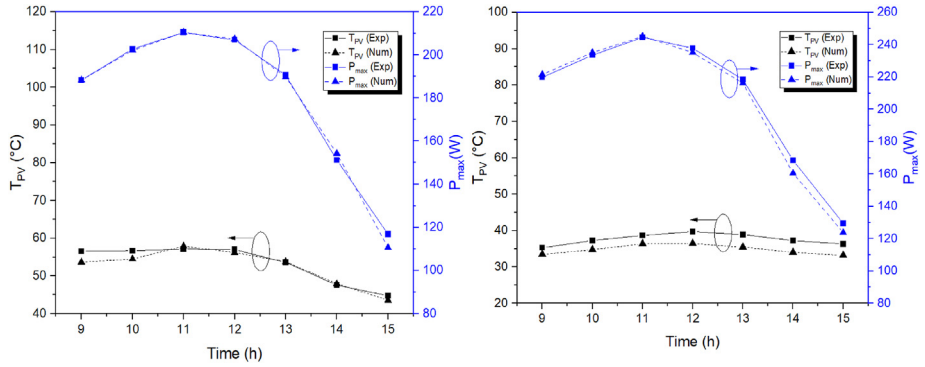



**Fig. 5.** Comparison between experimental and numerical results of the hourly change of PV temperature and maximum power for: a) no-cooling case and b) backside cooling case.

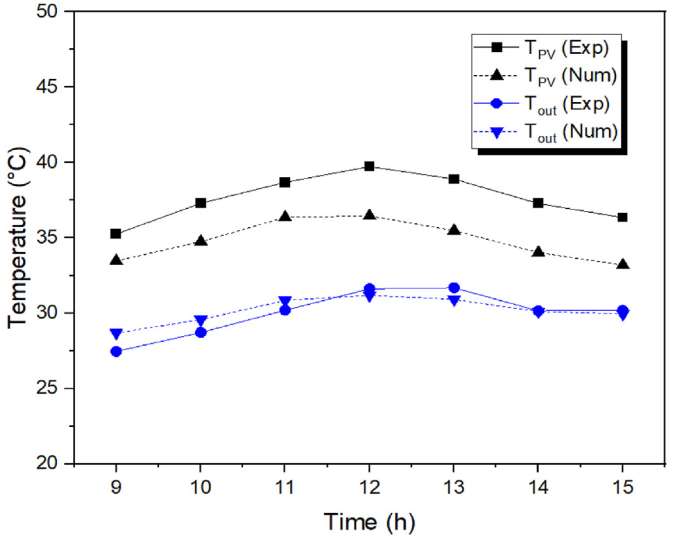



**Fig. 6.** Comparison between experimental and numerical results of the hourly change of the outlet fluid temperature for the hybrid PV-thermal system.

this PV-thermal module has a great impact on the electrical performance of the module due to the high thermal contact and the low thermal resistance between the solar cells and the heat exchanger to direct lamination.

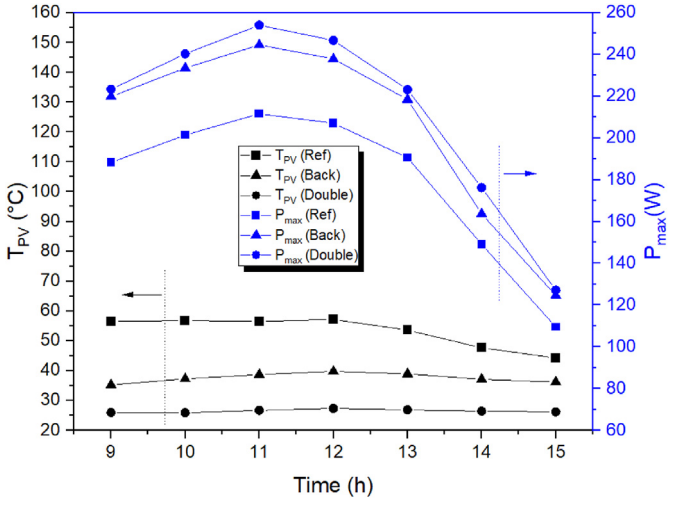



**Fig. 7.** Comparison between experimental data of the hourly change of the PV temperature and maximum power of the three solar modules.

In order to better understand the deviation of the electrical characteristics of the studied PV modules, the hourly change of $I_{sc}$ and $V_{oc}$ are compared in Fig. 9. The open circuit current follows the same profile as the solar radiation intensity. The short circuit current is almost the same for the studied modules and it is not affected by the operating temperature. However, the cooling effect of the PV is easily detected in the open circuit voltage behavior. A significant improvement of $V_{oc}$ is observed in the case of PV-thermal and double cooling methods, mainly due to the significant decrease of the cell temperature as shown in Fig. 7. In addition, the combined effect of variable solar radiation intensity and solar cell temperature effected the hourly variation of the $V_{oc}$. For instance, the decrease of the $V_{oc}$ of the module reference from 9:00–11:00 could be explained by the decrease on $V_{oc}$ due to the increase of cell temperature overcome the increase $V_{oc}$ due to the increase of solar radiation intensity. Moreover, the increase of $V_{oc}$ due to the decrease of the reference cell temperature from 11:00–15:00 has overcame the decrease of the $V_{oc}$ due to the decrease of the solar radiation intensity. The profile of $V_{oc}$ with active cooling was almost linear indicating that both effects (solar radiation intensity and PV) were similar except after 13:00 p.m. when the decrease of the $V_{oc}$ due to decrease of the solar radiation intensity was more pronounced than the temperature effect.

Based on the first law of thermodynamics, the innovative PV-

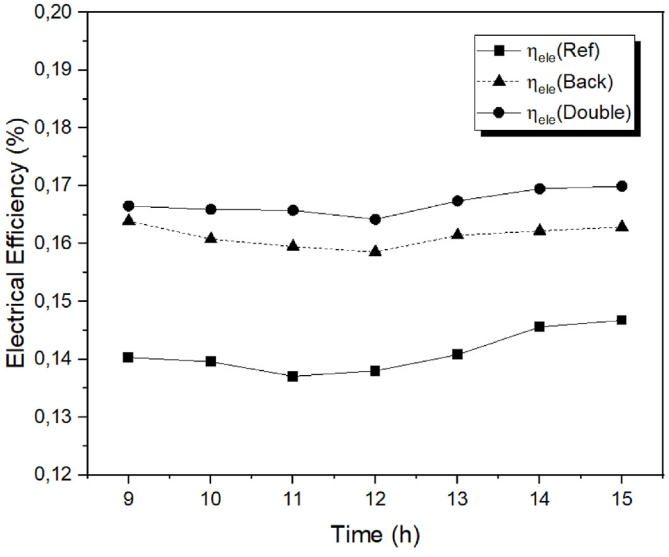



**Fig. 8.** Hourly variation of the measured electrical efficiency for the studied PV modules.

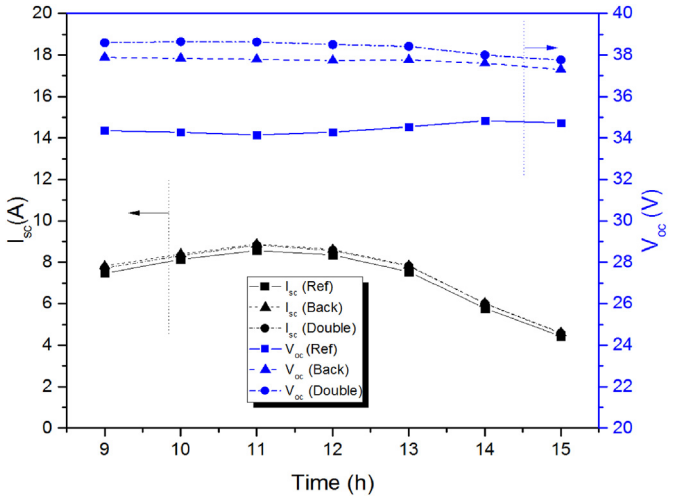



**Fig. 9.** Hourly variation of the $I_{sc}$ and $V_{oc}$ for the studied PV modules.

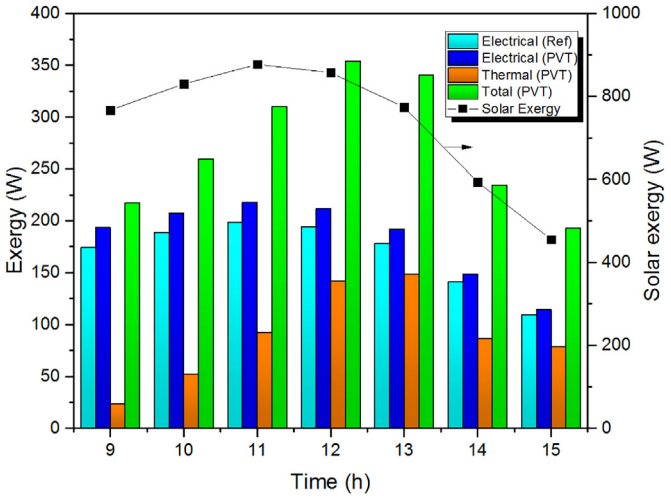



**Fig. 10.** Exergies variation of the PV-thermal system and un-cooled PV module.

thermal collector can generate greater electrical energy and extra thermal energy in respect to the reference module (uncooled). However, both generated energies by the novel system are not at the same thermodynamic quality, which should be compared based on the second law of thermodynamics. To this aim, an exergy analysis was carried out to evaluate the thermodynamic quality of the total energy harvested by the novel PV-thermal collector. Firstly, the exergy generated by the reference module is determined where the available energy coincide with the electrical energy. It is clear from Fig. 10 that exergy of the reference module follows the same trend of the solar irradiance exergy as the PV output power is proportional to the incident irradiance.

Secondly, the novel PV-thermal system's exergy content (electrical, thermal and total) is compared with the electrical exergy of the reference module in Fig. 10. The results showed that the electrical exergy of the innovative PV-thermal collector exceeded the reference module by an average of 8.5% because of the innovative cooling mechanism that reduced the solar cell temperature throughout the day. Besides this electrical exergy content, the PV-thermal system produced an extra thermal exergy that increased during noon time as the PV temperature increased. Although the thermal exergy production was lower than the electrical exergy, the total exergy of the PV-thermal showed a 61% enhancement compared to the reference PV. Thirdly, the exergy destruction and losses were estimated for uncooled and PV-thermal modules and compared in Fig. 11. The thermal management of the PV module led to a considerable reduction in the exergy losses. The back cooling system had the lowest exergy losses, which means that the cooling is a quite significant factor that could help to improve the overall yield of the PV systems. Exergy is always demolished when the system includes a temperature change. This destruction is proportional to the entropy generation within the system and its surroundings. The maximum exergy losses were observed at noontime at the high solar radiation, which heated the WF at low temperature levels. Hence, a formation of considerable irreversibility happened. Comparing the PV-thermal with the reference PV, a considerable reduction in the exergy destruction was detected as the PV reference converted the solar exergy only to electrical exergy and a bigger amount of the incident solar exergy was lost.

For the hybrid PV-thermal, a considerable reduction in exergy destruction was achieved when it compared with the reference case. The minimum and maximum reduction in exergy losses using the cooling system was about 7.25% at 9:00 and 27.3% at 12:00, respectively. Although the hybrid PV-thermal system achieved a considerable enhancement, more efforts should be made to optimize the useful exergy and harvest the maximum possible solar energy.

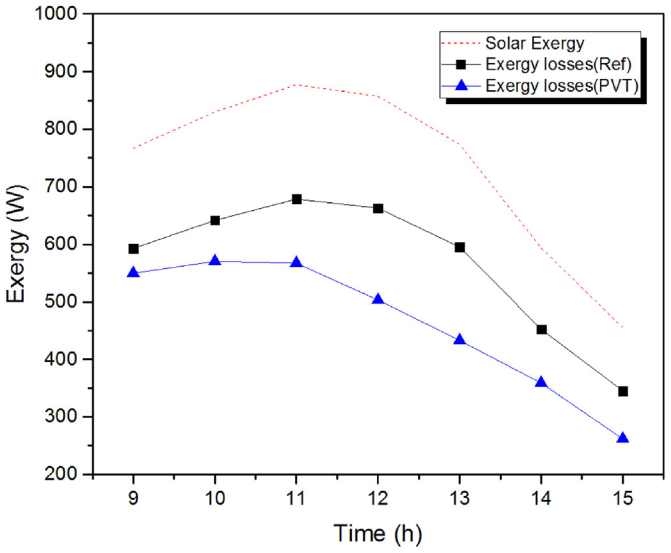



**Fig. 11.** Hourly exergy destruction of the three studied PV modules.

## 7. Conclusion

The current work presents a new thermal management system for an innovative PV-thermal collector. The performance of the new system was comprehensively investigated both numerically and experimentally. An experimental installation formed by mechanical, hydraulic and electrical components is designed and tested under real weather conditions of Sharjah, UAE. The reliability of the measurement data was checked using an uncertainty analysis. Moreover, a numerical model based on thermal resistance network was developed to predict the electrical and thermal performances of the PV-thermal collector. The calculated results showed a good agreement with experimental data for both cases with and without cooling. In addition, an exergy analysis was carried out to assess the effectiveness of the active cooling using the innovative heat exchanger from the second law of thermodynamic perspective. Furthermore, the electrical characteristics of the studied modules

were investigated. The variation of the short circuit current followed the solar radiation intensity. Further, the results showed that the cooling effect of the solar cell has contributed to increase the open circuit voltage respect the uncooled case. The variation of the open circuit voltage versus time was influenced by the combination of the variable solar radiation intensity as well as the change in the solar cell temperature.

The main research findings and results are summarized as following:

- A significant enhancement in power output of about 15% was achieved with the PV-thermal with the new backside cooling system.
- The average module temperature was reduced to 16 °C in case of the backside cooling compared to the non-cooling case. Consequently, the electrical efficiency improved up to 14.3%
- Including the front cooling to the PV-thermal system has contributed to an additional cooling of the solar cell whose temperature decreased by an average of 26 °C respect to the reference case.
- An increase of 18.3% in electrical efficiency due to the double cooling (front and backside) method was reported.
- For the front cooling method, other challenges related to the control and thermal stress on the upper glass surface should be also considered.
- The energy improvement using the novel PV-thermal, results on exergy analysis demonstrated that the overall exergy raised up to 61% compared to the reference module. This reduction in the exergy losses occurred because of the efficient thermal management. The minimum and maximum reduction in exergy losses within the cooled system were about 7.25% (at 9:00) to 27.3% (at noon), respectively.
- The rejected heat from the novel system cloud be used for various domestic heating applications.

The results of this work could reveal the electrical, thermal, and thermodynamic characteristics of the innovative PV-thermal system under UAE weather conditions. It is a first test with different cooling methods under real operating conditions. Future works will be conducted to evaluate other cooling methods such as the use of nanofluids and phase change materials. It is also of great interest to investigate the thermo-hydraulic performance of the innovative PV-thermal system and compare it with other designs of PV-thermal systems.

## CRediT authorship contribution statement

**Ahmed Amine Hachicha:** Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Essam M. Abo-Zahhad:** Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Zafar Said:** Methodology, Investigation, Writing – review & editing. **S.M.A. Rahman:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Nomenclature

*Roman letters*

| | |
|---|---|
| A | area [$m^2$] |
| e | root mean square percent deviation |
| E | exergy [W] |
| F' | collector efficiency factor |
| G | solar irradiance [$Wm^{-2}$] |
| h | convection heat transfer coefficient [$Wm^{-2}K^{-1}$] |
| I | current [A] |
| k | thermal conductivity [$Wm^{-1}K^{-1}$] |
| L | length (m) |
| Q | heat flux transfer [$Wm^{-2}$] |
| r | coefficient of correlation |
| R | thermal resistance [K $W^{-1}$] |
| S | entropy [J $kg^{-1}\cdot K^{-1}$] |
| T | temperature [° C] |
| U | overall heat transfer coefficient [$Wm^{-2}K^{-1}$] |
| V | voltage [V] |

*Greek letters*

| | |
|---|---|
| α | absorptivity |
| β | cell temperature coefficient ($K^{-1}$) |
| Δ | difference |
| τ | transmittivity |
| ω | uncertainty |

*Subscripts*

| | |
|---|---|
| cond | conduction |
| conv | convection |
| el | electrical |
| exp | experimental |
| f | fluid |
| g | glass |
| oc | open circuit |
| rad | radiation |
| ref | reference |
| sc | short circuit |
| sim | simulation |
| th | thermal |
| tot | total |

*Abbreviations*

| | |
|---|---|
| CFD | Computational Fluid Dynamics |
| EVA | Ethylene-vinyl acetate |
| Pmax | Maximum Power |
| PV | thermal Photovoltaic-thermal collector |
| Impp | Maximum power point current |
| Isc | Short circuit current |
| MPPT | Maximum power point tracker |
| WF | Working fluid |
| ST | Solar thermal |
| $V_{mpp}$ | Maximum power pointvoltage |
| Voc | Open circuit voltage |

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